

1. Administrative & Study Identification

Full Study Title: A Phase 2a Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Efinopegdutide (MK-6024) in Adults With Compensated Cirrhosis Secondary to Metabolic Dysfunction-Associated Steatohepatitis
Short Title/Acronym: MK-6024-017 (Efinopegdutide in MASH Cirrhosis) **Protocol Number:** MK-6024-017 **NCT Number:** NCT06465186
Sponsor Name: Merck Sharp & Dohme LLC **Investigational Product:** Efinopegdutide (MK-6024) **Phase of Development:** Phase 2a **Medical Monitor:** Clinical Director, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if efinopegdutide can lower the amount of fat, inflammation, and scarring (fibrosis) in the liver in adults with compensated cirrhosis secondary to MASH. **Secondary Objectives:** To evaluate the safety and tolerability of efinopegdutide compared to placebo, and to assess its effects on body weight and non-invasive markers of liver stiffness.

2.2 Background & Rationale Metabolic dysfunction-associated steatohepatitis (MASH, previously known as NASH) is a severe form of fatty liver disease linked to obesity, type 2 diabetes, and metabolic syndrome. MASH leads to a buildup of fat and inflammation, which can progress to compensated cirrhosis (scarring of the liver), liver failure, and cancer. There remains a significant unmet medical need for treatments that can halt or reverse cirrhosis in MASH patients. Efinopegdutide is an investigational medication being studied to determine if it can effectively reduce liver fat, resolve inflammation, and improve liver fibrosis safely.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 80 evaluable patients **Number of Sites:** 69 locations globally
Estimated Study Start: 12-Jul-2024 **Estimated Primary Completion:** 06-Aug-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 80 years. 2. Confirmed diagnosis of compensated cirrhosis caused by metabolic dysfunction-associated steatohepatitis (MASH). 3. Has either type 2 diabetes that is controlled by diet or medication, or does not have type 2 diabetes.

4.2 Exclusion Criteria 1. History of a liver disease other than MASH (e.g., Hepatitis B or C, drug-induced liver disease, or autoimmune liver disease). 2. History of type 1 diabetes or history of pancreatitis. 3. Had a bariatric surgical procedure less than 5 years before entry into the study. 4. Major illnesses such as recent (within 6 months of study entry) episodes of heart problems, including congestive heart failure, unstable angina, heart attack, stroke, or mini-stroke.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Weekly administration. Participants start at the lowest dose level, escalating every month for three months until reaching the highest dose level. **Route of Administration:** Subcutaneous (SC) injection via a prefilled syringe. **Duration of Treatment:** 28 weeks (approx. 7 months), with a total study duration of ~43 weeks (about 10 months).

5.2 Concomitant & Prohibited Therapy Permitted: Standard medications for the management of controlled type 2 diabetes and associated metabolic comorbidities. **Prohibited:** Concomitant use of other investigational weight-loss medications or other experimental treatments for liver disease.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Liver Imaging/Biopsy evaluation
Baseline	Day 0	Randomization, First Dose (Lowest level), Safety Labs
Interim	Weeks 4 to 24	Monthly Dose Escalation, Safety Labs, Efficacy Assessment, Body Weight Checks
End of Study	Week 28	Final Efficacy Assessment (Liver fat/inflammation/fibrosis imaging), IP Accountability
Follow-up	Up to Week 43	Safety and tolerability follow-up phone calls/visits

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Improvement in liver histology, specifically evaluating the reduction in the amount of liver fat, inflammation, and scarring (fibrosis) at Week 28. **Measurement**

Method: Central Laboratory Analysis / Liver imaging tests (e.g., MRI-PDFF) and non-invasive fibrosis scoring.

7.2 Secondary & Exploratory Endpoints 1. Safety and tolerability of efinopegdutide (incidence of AEs/SAEs). 2. Change from baseline in body weight. 3. Change from baseline in non-invasive markers of liver stiffness/fibrosis (e.g., ELF score, cT1, Pro-C3, FIB-4, VCTE).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collected via physical examinations, blood samples, and patient questionnaires across approximately 10 scheduled onsite visits and at least 3 phone calls. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours of site awareness). **Data Monitoring Committee (DMC):** Sponsor-designated safety review committee.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for all tolerability and AE analyses. **Sample Size Justification:** Target $N = 80$ appropriately powered to detect meaningful shifts in liver fat, inflammation, and fibrosis staging compared to placebo. **Handling of Missing Data:** Mixed-effect Model Repeated Measure (MMRM) and/or Multiple Imputation for continuous endpoints (e.g., body weight, liver fat content).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring across the 69 global trial locations. **Audit Trail:** Secure eCRF entry, locking mechanisms, and centralized reading for liver imaging and biomarker assays.

11. Study Identifiers & Governance

Primary ID: MK-6024-017 **Registry Number:** NCT06465186 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** MK-6024-017 Efinopegdutide in MASH Cirrhosis **Official Regulatory Title:** A Phase 2a Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Efinopegdutide (MK-6024) in Adults With Compensated Cirrhosis Secondary to Metabolic Dysfunction-Associated Steatohepatitis

12. Clinical Framework (MASH Specific)

Primary Condition: Compensated Cirrhosis Secondary to MASH
Diagnosis Threshold: Liver scarring/damage caused by MASH confirmed by imaging/biopsy
Medical Methodology: Investigational targeted therapy (Efinopegdutide vs. Placebo)
Optimization Target: Reduction in liver fat, inflammation, and fibrosis staging alongside body weight management

13. Study Procedures & Methodology

Management Pathway: Weekly subcutaneous (SC) injections with monthly dose escalation over 3 months
Education Protocol (HCP): Protocol training, SC prefilled syringe administration, and AE monitoring
Education Protocol (Patient): SC injection technique instruction, trial expectations, and lifestyle/dietary counseling
Quality Audit Mechanism: Centralized independent review of liver imaging and routine clinical lab verification

14. Observation & Follow-Up Schedule

Total Duration: 43 weeks (approximately 10 months)
Onsite Visit Cadence: Around 10 scheduled visits for medical tests and assessments
Remote Monitoring Frequency: At least 3 phone call visits with the study doctor/team
Checklist Review Interval: Every 4 weeks (Q4W) during the 3-month dose escalation phase

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				